CONTROL SYSTEMS

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CONTROL SYSTEMS

Unit 1: Mathematical Models Of Physical Systems

CONTROL SYSTEMS

ANALOGOUS SYSTEMS

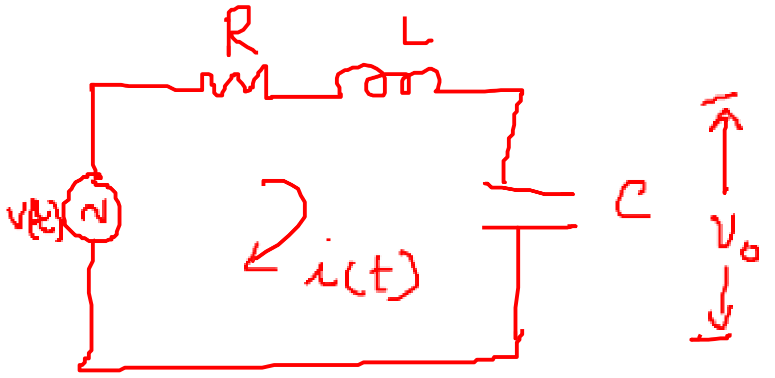
CONTROL SYSTEMS

FORCE-VOLTAGE ANALOGOUS SYSTEMS

Mathematical Models Of Physical Systems

Example

- Simple System, RLC Circuit



KVL,

$$v(t) = R\,i(t) + L\frac{di}{dt} + \overbrace{\frac{1}{C}\int_{-\infty}^{t} i(\tau)\,d\tau}^{v_o(t)} \rightarrow (1)$$

$v(t) = V$

$$v_o(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)\,d\tau$$

$$V_o(s) = \frac{I(s)}{Cs}$$

→ Integro-differential equation

(1) diff. w.r.to t

$$0 = R\frac{di}{dt} + L\frac{d^2i}{dt^2} + \frac{i(t)}{C}$$

$$\Rightarrow L\frac{d^2i}{dt^2} + R\frac{di}{dt} + \frac{i(t)}{C} = 0 \rightarrow (2)$$

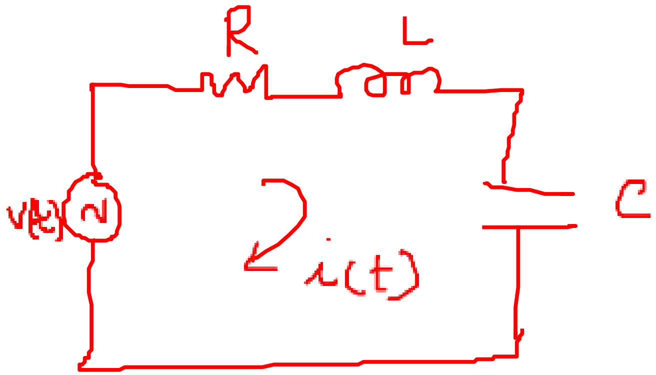
If order is higher ⇒

⇒ Applying LT (2) ⇒ $L s^2 I(s) + R s I(s) + \dfrac{I(s)}{Cs} = 0$

Mathematical Models Of Physical Systems

Example

- Simple System, RLC Circuit



① LT

$$V(s) = \left(R + Ls + \frac{1}{Cs}\right) I(s)$$

$$\frac{V_0(s)}{V(s)} = \frac{\frac{I(s)}{Cs}}{\left(R + Ls + \frac{1}{Cs}\right) I(s)}$$

$$= \frac{1}{RCs + LCs^2 + 1}$$

→ mathematical model

$$\frac{I(s)}{V(s)} = \frac{Cs}{LCs^2 + RCs + 1}$$

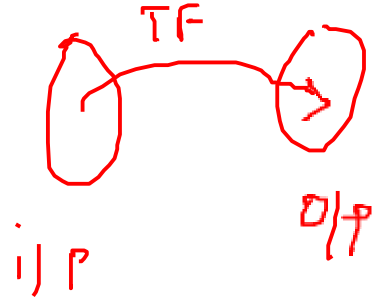
→ algebraic function

Mathematical Models Of Physical Systems

Example

- Simple System, Mass Spring Damper

ex. modelling dynamics of car



→ state space ⇒ higher order s/m

↓

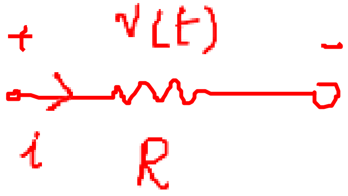
convert this into simple first order s/m

$$\frac{dx}{dt} = -ax + b$$ → ro diff.

Mathematical Models Of Physical Systems

Idealizing Assumptions

- Linear: The components are said to be linear if they follow super position principle.



$\Rightarrow v(t) = R\,i(t) \rightarrow$ ohm's law is true under certain idealizing assumptions

ex. tunnel diode

$u \rightarrow \boxed{\text{System}} \rightarrow y$

$u_1 \longrightarrow y_1$

$u_2 \longrightarrow y_2$

$u_1 + u_2 \longrightarrow y_1 + y_2 \rightarrow$ additivity

$\alpha u \longrightarrow \alpha y \rightarrow$ homogeneity

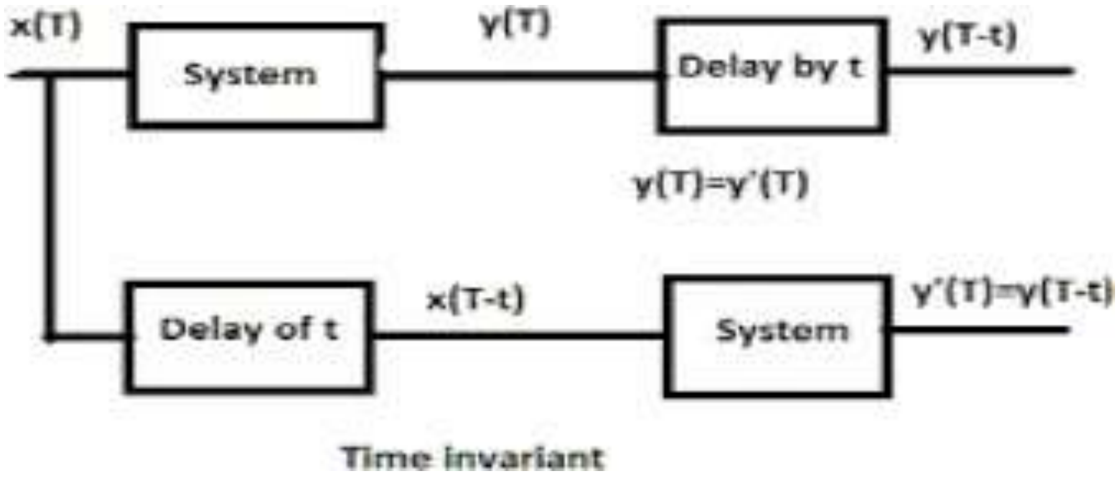
} principle of superposition

↳ linearity

Idealizing Assumptions

- Time Invariant:

This means the behavior of **system** does not depend on **time** at which input is applied.



MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

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Differential Equations Of Physical Systems

Mechanical Systems

The differential equation modeling of mechanical systems.
There are two types of mechanical systems based on the type of motion:

- Translational mechanical systems
- Rotational mechanical systems

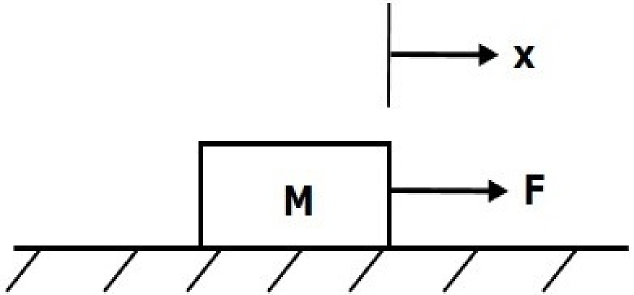
Modeling of Translational Mechanical Systems

- Translational mechanical systems move along a **straight line**.
- These systems mainly consist of three basic elements. Those are mass, spring and dashpot or damper.

Differential Equations of Physical Systems
Mechanical Systems

Mass

Mass is the property of a body, which stores **kinetic energy**. If a force is applied on a body having mass **M**, then it is opposed by an opposing force due to mass. This opposing force is proportional to the acceleration of the body. Assume elasticity and friction are negligible.



$$F_m \propto a$$

$$\Rightarrow F_m = Ma = M\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$$

$$F = F_m = M\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$$

Where,

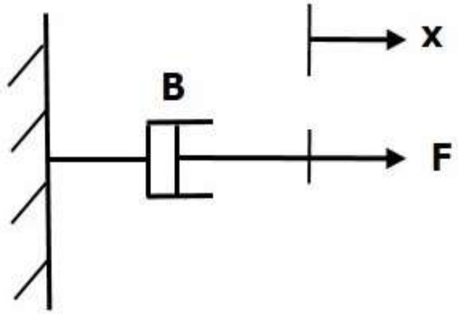
- **F** is the applied force
- $\mathbf{F_m}$ is the opposing force due to mass
- **M** is mass
- **a** is acceleration
- **x** is displacement

Differential Equations of Physical Systems

Mechanical Systems

Dashpot

If a force is applied on dashpot **B**, then it is opposed by an opposing force due to **friction** of the dashpot. This opposing force is proportional to the velocity of the body. Assume mass and elasticity are negligible.



$$F_b \propto \nu$$

$$\Rightarrow F_b = B\nu = B\frac{\mathrm{d}x}{\mathrm{d}t}$$

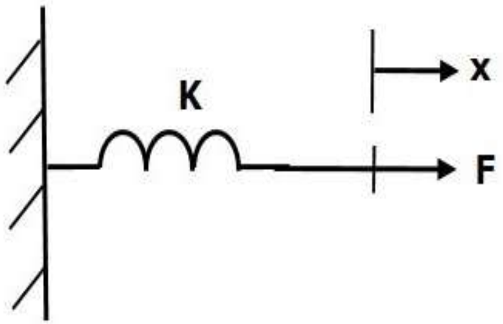
$$F = F_b = B\frac{\mathrm{d}x}{\mathrm{d}t}$$

Where,

- $\mathbf{F_b}$ is the opposing force due to friction of dashpot
- **B** is the frictional coefficient
- **v** is velocity
- **x** is displacement

Spring

Spring is an element, which stores **potential energy**. If a force is applied on spring **K**, then it is opposed by an opposing force due to elasticity of spring. This opposing force is proportional to the displacement of the spring. Assume mass and friction are negligible.



$$F \propto x$$

$$\Rightarrow F_k = Kx$$

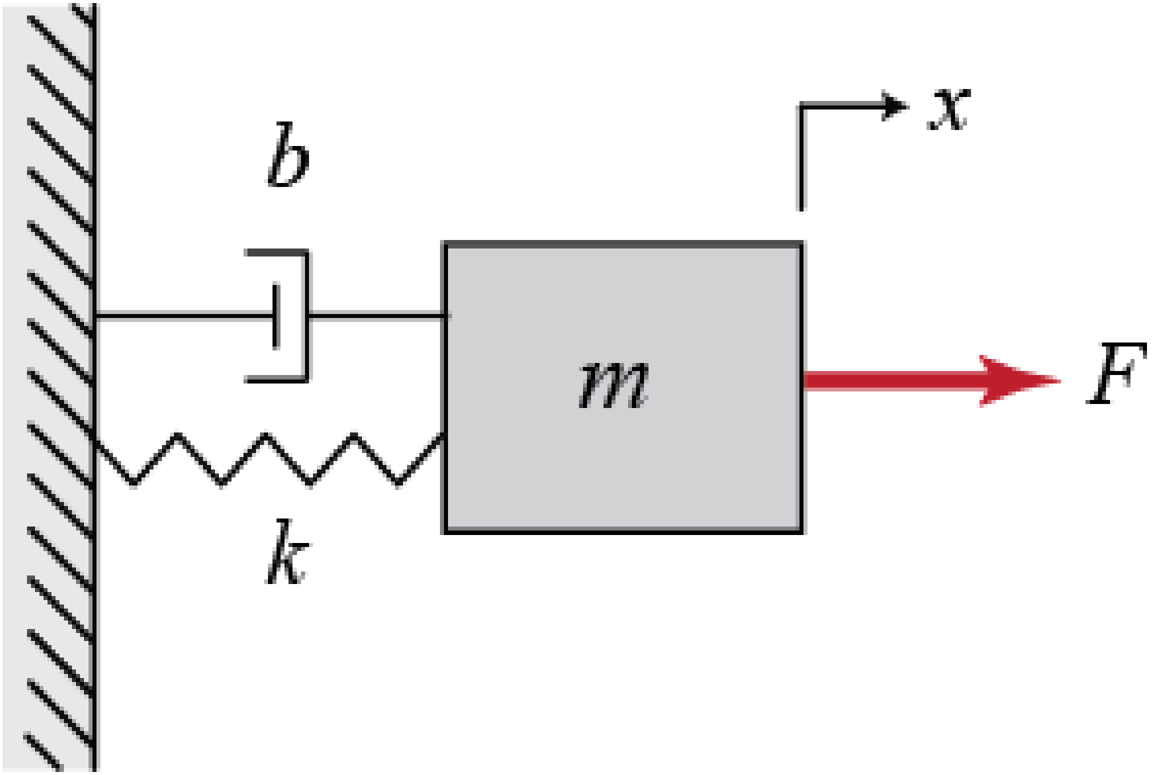
$$F = F_k = Kx$$

Where,

- **F** is the applied force
- $\mathbf{F_k}$ is the opposing force due to elasticity of spring
- **K** is spring constant
- **x** is displacement

Differential Equations of Physical Systems

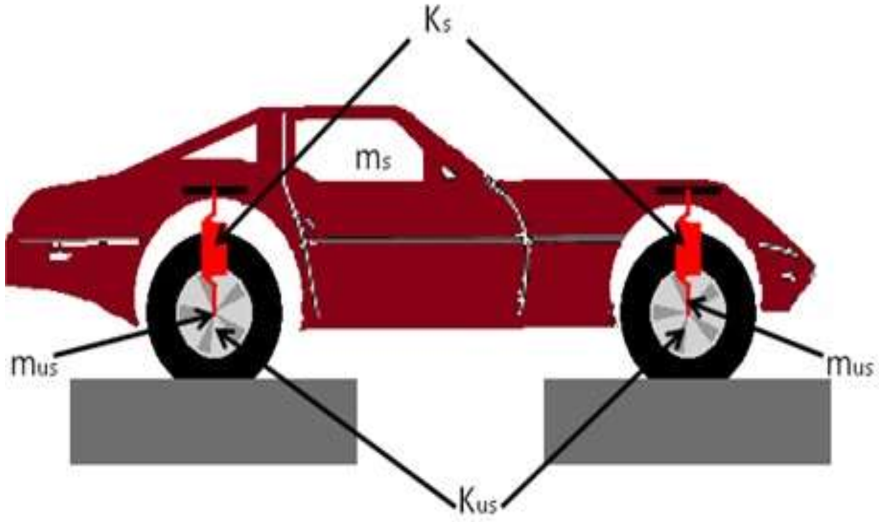
Mechanical Systems

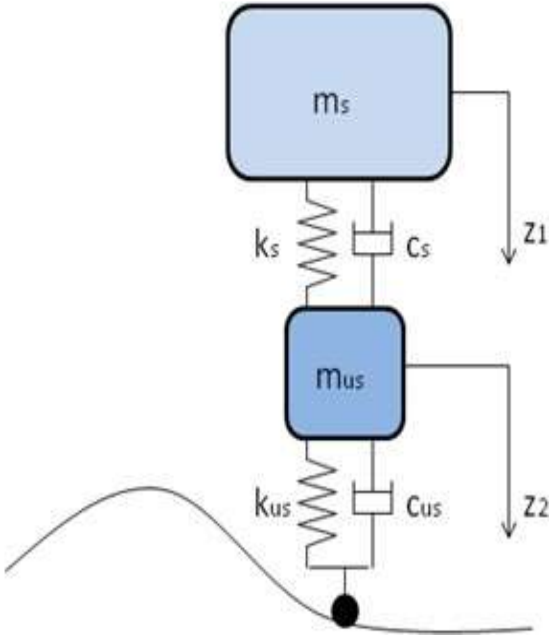


Differential Equations of Physical Systems

Mechanical Systems

- Systems that consist of spring , mass and damper. Ex, modeling of a car

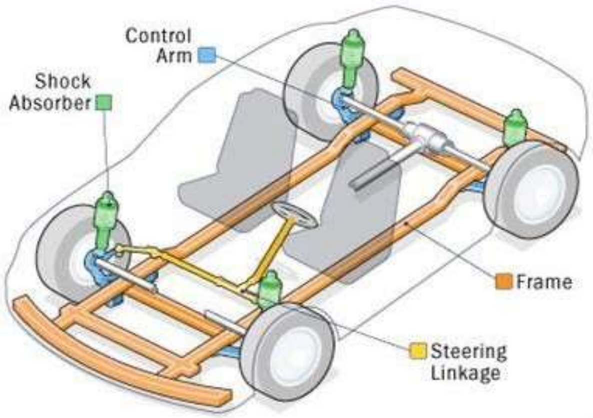


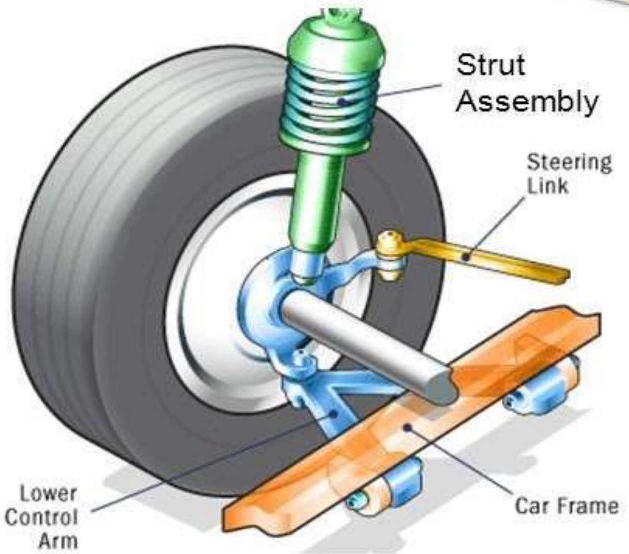


Differential Equations of Physical Systems

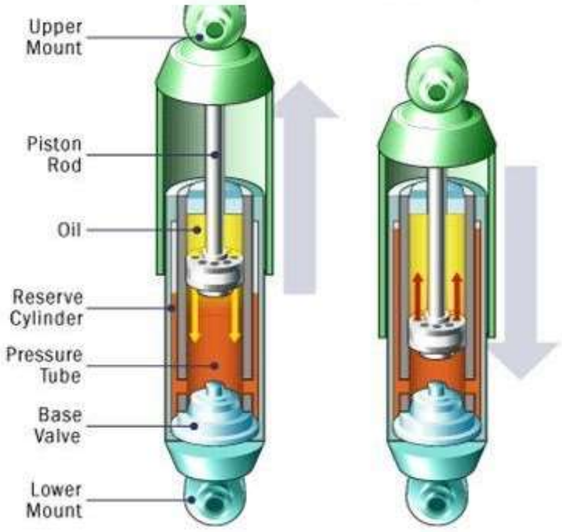
Mechanical Systems

The Suspension System



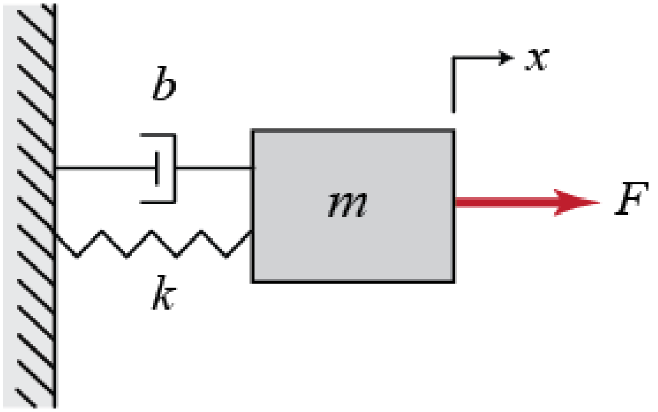


Dampers



- Systems that consist of spring , mass and damper.
- Force exerted in each element by considering x as displacement and F is force ,B or f is viscous damper, k is spring constant and M is mass as per D'Alemberts principle is:
- $F = F_m + F_b + F_k$

$$F = M\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx$$



Converting a mechanical system to electrical system

There are 2 ways :

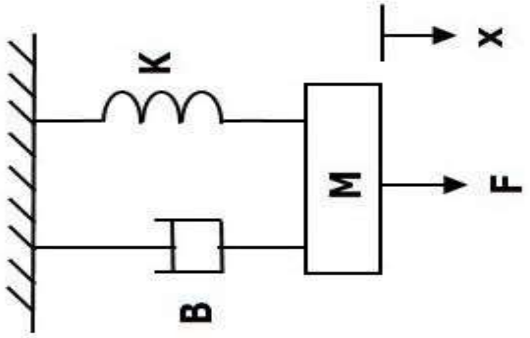
Force – Current Analogy

- In this case , force is analogous to current and velocity is analogous to voltage.

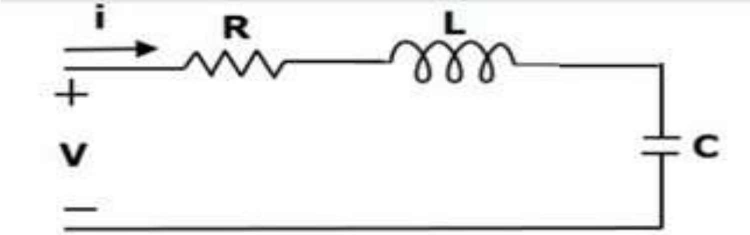
Force –Voltage Analogy

- In this case , force is analogous to voltage and velocity is analogous to current.

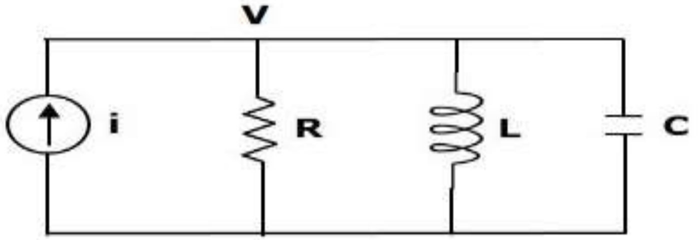
Converting a Translational Mechanical System to Electrical



$$F = M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + Kx \quad (1)$$



$$V = L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q \quad (2)$$



$$i = C\frac{d^2\psi}{dt^2} + \frac{1}{R}\frac{d\psi}{dt} + \frac{\psi}{L} \quad (3)$$

where ψ= flux linkages

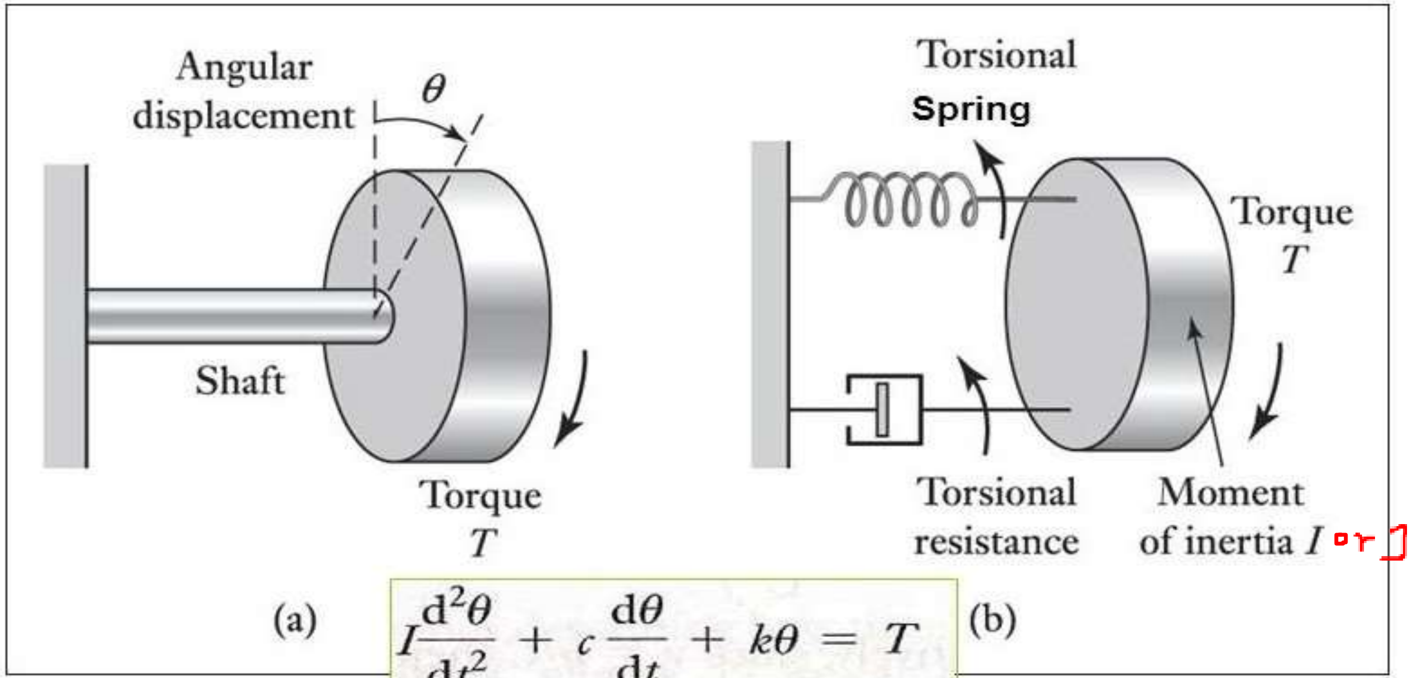
Mechanical system	Force-voltage	Force-current
Force(F)	Voltage(V)	Current(i)
Mass(M)	Inductance(L)	Capacitance(C)
Frictional Coefficient(B)	Resistance(R)	Reciprocal of Resistance($\frac{1}{R}$)
Spring Constant(K)	Reciprocal of Capacitance ($\frac{1}{c}$)	Reciprocal of Inductance($\frac{1}{L}$)
Displacement(x)	Charge(q)	Magnetic flux(ψ)
Velocity(v)	Current(i)	Voltage(V)

Differential Equations of Physical Systems

Examples of Mechanical Systems - Rotational

- Rotational : I or J as moment of inertia, c or B as damping constant, T as Torque

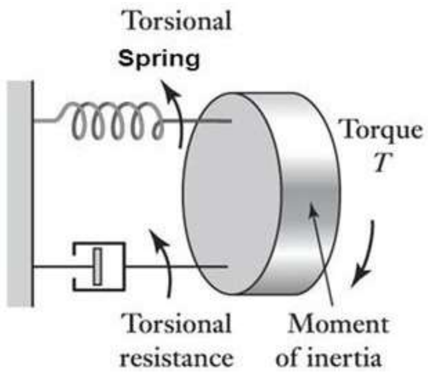
D

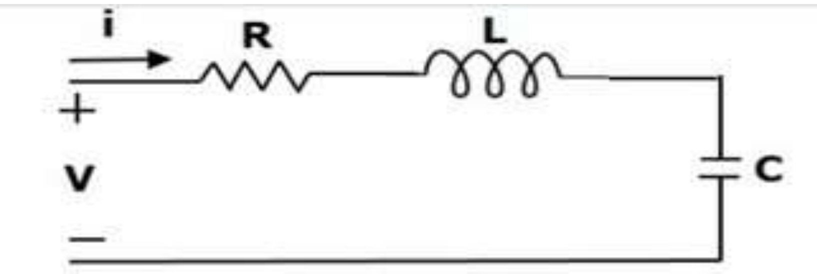


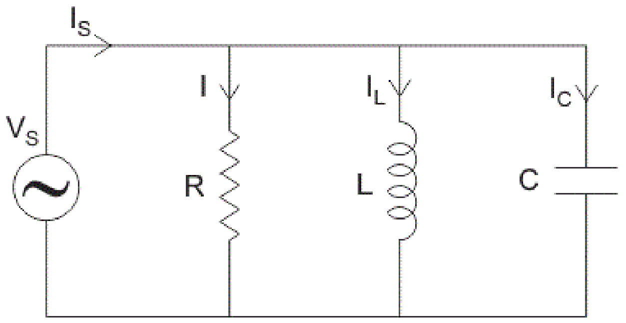
$$I\frac{d^2\theta}{dt^2} + c\frac{d\theta}{dt} + k\theta = T$$

Rotating a mass on the end of a shaft: (a) physical situation, (b) building block model

Converting a mechanical system to electrical system







$$T = J\frac{d^2\theta}{dt^2} + B\frac{d\theta}{dt} + K\theta$$

$$V = L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q$$

$$i = C\frac{d^2\psi}{dt^2} + \frac{1}{R}\frac{d\,\psi}{dt} + \frac{\psi}{L}$$

Rotational Mechanical System	Electrical System
Torque(T)	Voltage(V)
Moment of inertia(J)	Inductance(L)
Rotational friction coefficient(B)	Resistance(R)
Torsional spring constant(K)	Reciprocal of Capacitance ($\frac{1}{c}$)
Angular displacement(θ)	Charge(q)
Angular velocity(ω)	Current(i)

Electrical System
Current(i)
Capacitance(C)
Reciprocal of Resistance($\frac{1}{R}$)
Reciprocal of Inductance($\frac{1}{L}$)
Magnetic Flux(ψ)
Voltage(V)

Definition of Laplace Transform

$$F(s) = L\{f(t)\} = \int_{0^-}^{\infty} f(t)e^{-st}dt$$

Sufficient conditions for existence of the Laplace transform

- $f(t)$ is piecewise continuous
- There exist M, α, t_0 such that

$$|f(t)| < M\ e^{\alpha t} \quad for \quad t \geq t_0$$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Laplace Transform Table

$f(t)$	$\mathcal{L}[f(t)]$
1	$\frac{1}{s}$
e^{at}	$\frac{1}{s-a}$
$\sin at$	$\frac{a}{s^2+a^2}$
$\cos at$	$\frac{s}{s^2+a^2}$
$t \sin at$	$\frac{2as}{(s^2+a^2)^2}$
$t \cos at$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
$e^{at} \sin bt$	$\frac{b}{(s-a)^2+b^2}$
$e^{at} \cos bt$	$\frac{s-a}{(s-a)^2+b^2}$
$\frac{t^n}{n!}, n \in \mathbb{N}$	$\frac{1}{s^{n+1}}$
$e^{at} \cdot \frac{t^n}{n!}, n \in \mathbb{N}$	$\frac{1}{(s-a)^{n+1}}$

For first-order derivative:

$$\mathcal{L}\{f'(t)\} = s\,\mathcal{L}\{f(t)\} - f(0)$$

For second-order derivative:

$$\mathcal{L}\{f''(t)\} = s^2\mathcal{L}\{f(t)\} - s\,f(0) - f'(0)$$

For third-order derivative:

$$\mathcal{L}\{f'''(t)\} = s^3\mathcal{L}\{f(t)\} - s^2 f(0) - s\,f'(0) - f''(0)$$

For n$^{\text{th}}$ order derivative:

$$\boxed{\mathcal{L}\{f^n(t)\} = s^n\mathcal{L}\{f(t)\} - s^{n-1} f(0) - s^{n-2} f'(0) - \cdots - f^{n-1}(0)}$$

Proof of Laplace Transform of Derivatives

$$\mathcal{L}\{f'(t)\} = \int_0^\infty e^{-st} f'(t)\,dt$$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions

- A control system consists of an output as well as an input signal. The output is related to the input through a function called **transfer function**.

excitation
cause

effect

$R(s)$ $G(s)$ $C(s) = G(s)\,R(s) \Rightarrow \dfrac{C(s)}{R(s)} = G(s)$

- In Laplace Transform, if the input/cause is represented by R(s) and output/effect is represented by C(s), then the transfer function will be
- $$C(S) = G(S)R(S)$$
- The **transfer function** of a control system is defined as the ratio of the Laplace transform of the output variable to Laplace transform of the input variable assuming **all initial conditions to be zero.**

$$\frac{C(s)}{R(s)}$$

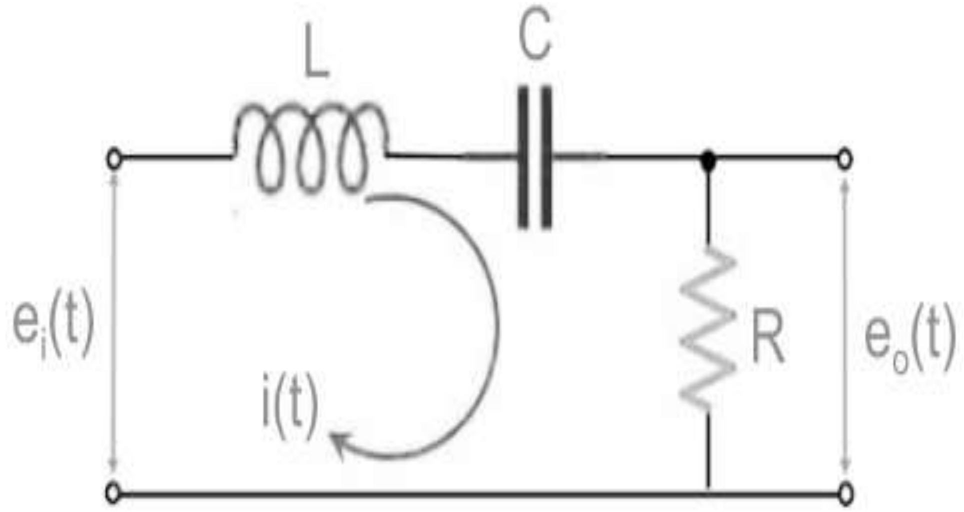
$\uparrow$

$f(0),\ f'(0),\ f''(0),\ \ldots$

- For any control system there exists a reference input termed as **excitation or cause** which operates through a transfer operation and produces an **effect** resulting in controlled **output** or response.

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions



By applying Kirchhoff's Voltage Law (KVL) to the above circuit, we get:

$$e_i\left(t\right) = L\frac{di(t)}{dt} + \frac{1}{C}\int i\left(t\right)dt + i\left(t\right)R\ldots\ldots Eq(1)$$

And,

$$e_o\left(t\right) = i\left(t\right)R\ldots\ldots Eq(2)$$

Further, by neglecting the initial conditions and taking the Laplace transform of the above equations, we obtain:

$$E_i\left(s\right) = sLI(s) + \frac{1}{C}\frac{I(s)}{s} + RI\left(s\right)\ldots\ldots Eq\left(3\right)$$

$$E_o\left(s\right) = RI\left(s\right)\ldots\ldots\ldots Eq(4)$$

So,

$$E_i\left(s\right) = I\left(s\right)\left[sL + R + \frac{1}{sC}\right]\ldots\ldots Eq(5)$$

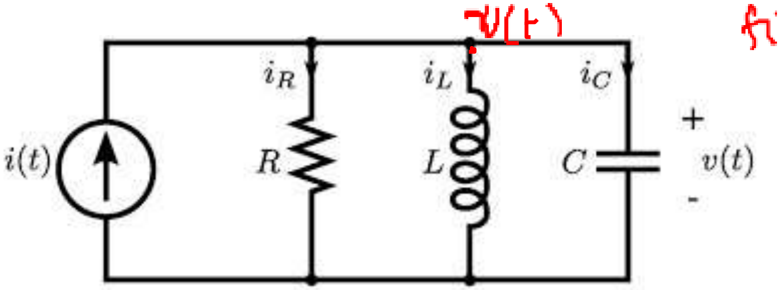
MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions Examples – H.W

- **Electrical systems:**

i/p | cause $= I(s)$, o/p | effect $= V(s)$

find T.F $\frac{V(s)}{I(s)}$



$$v(t) = \frac{1}{C}\int_{-\infty}^{t} i_c(\tau)\,d\tau$$

$$i(t) = \frac{v(t)}{R} + \frac{1}{L}\int_{-\infty}^{t} v(\tau)\,d\tau + C\frac{dv}{dt}$$

Apply LT with zero initial conditions

$$I(s) = \frac{V(s)}{R} + \frac{V(s)}{Ls} + CsV(s)$$

$$= \left(\frac{1}{R} + \frac{1}{Ls} + Cs\right)V(s)$$

$$\Rightarrow \frac{V(s)}{I(s)} = \frac{1}{\frac{1}{R} + \frac{1}{Ls} + Cs} = \frac{RLs}{Ls + R + LCRs^2}$$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions Examples

energy storing elements

electrical → L & C

mechanical → K & M

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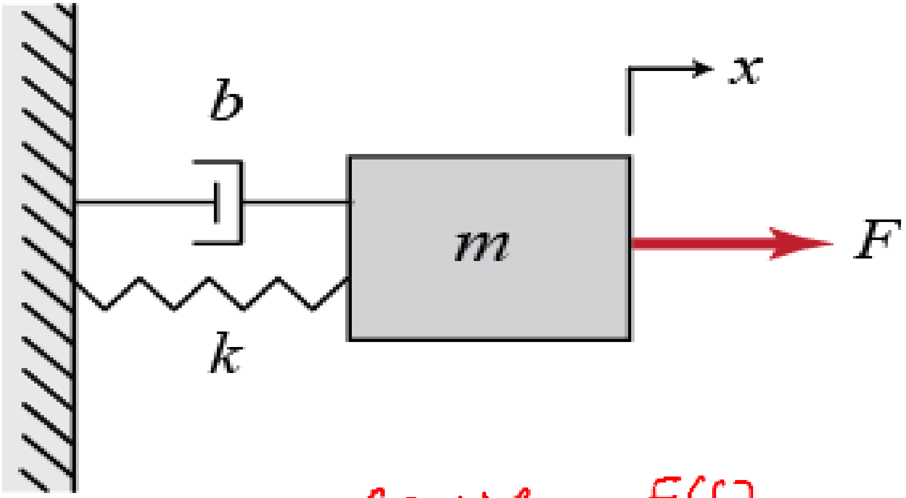
- Cause – Force and Effect – displacement x



$$F = m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx$$

$$= m\ddot{x} + b\dot{x} + kx$$

Apply LT with zero i.c

$$F(s) = m s^2 X(s) + b s X(s) + k X(s)$$

$$= (ms^2 + bs + k) X(s)$$

Cause = $F(s)$

effect = $X(s)$

Find T F $\frac{X(s)}{F(s)}$

$$\frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k} = \frac{N(s)}{D(s)} = \frac{\text{zero polynomial}}{\text{poles polynomial}}$$

roots

⇒ order of the system = highest degree of $D(s)$

⇒ 2 finite poles

2 zeros at ∞

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS
Transfer Functions Examples

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$

leading coefficient $= 1$

$\Rightarrow$ monic polynomial

$\deg N(s) \leq \deg D(s)$

$n \geq m$

$\Rightarrow$ strictly proper transfer function

- Cause – Force and Effect – displacement y_1 $\quad \frac{Y_1(s)}{F(s)}$

$$F(t) = M_1 \frac{d^2 y_1}{dt^2} + K_1 y_1 + b \frac{dy_1}{dt} + K_{12}(y_1 - y_2)$$

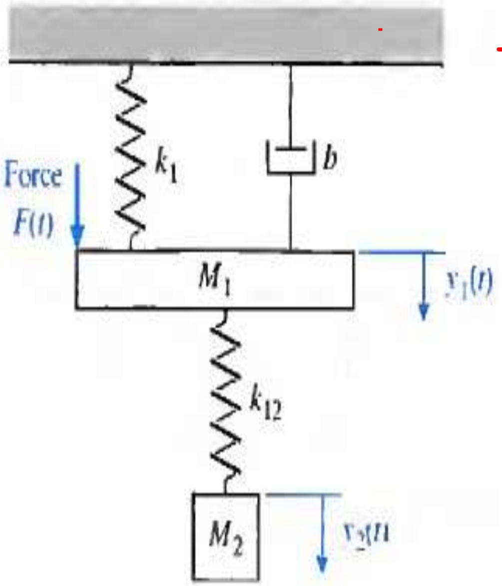
$$0 = M_2 \frac{d^2 y_2}{dt^2} + K_{12}(y_2 - y_1)$$

Apply LT with zero i c

$$F(s) = (M_1 s^2 + bs + K_1 + K_{12}) Y_1(s) - K_{12} Y_2(s)$$

$$0 = (M_2 s^2 + K_{12}) Y_2(s) - K_{12} Y_1(s)$$

$$\frac{Y_1(s)}{F(s)}$$



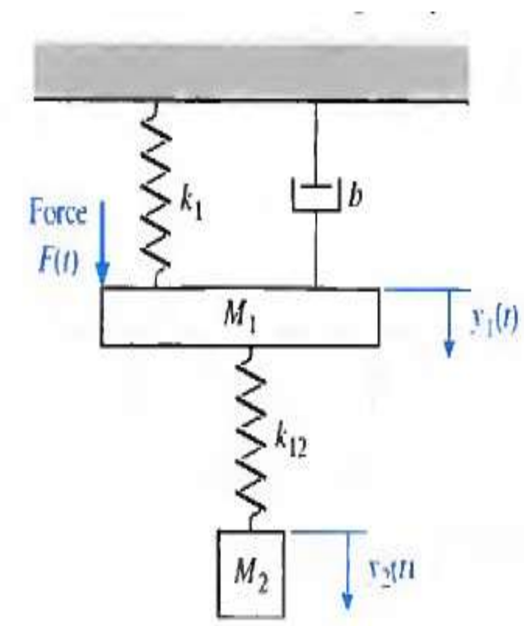
- Cause – Force and Effect – displacement y_1

$$M_1\ddot{y}_1 + k_{12}(y_1 - y_2) + b\dot{y}_1 + k_1 y_1 = F(t)$$

$$M_2\ddot{y}_2 + k_{12}(y_2 - y_1) = 0\,.$$

$\dot{y}_1 \rightarrow \frac{dy_1}{dt}$

$\frac{d^2y_1}{dt^2} = \ddot{y}_1$



$$(M_1 s^2 + k_{12} + bs + k_1)Y_1(s) - \frac{k_{12}k_{12}Y_1(s)}{M_2 s^2 + k_{12}} = F(s)$$

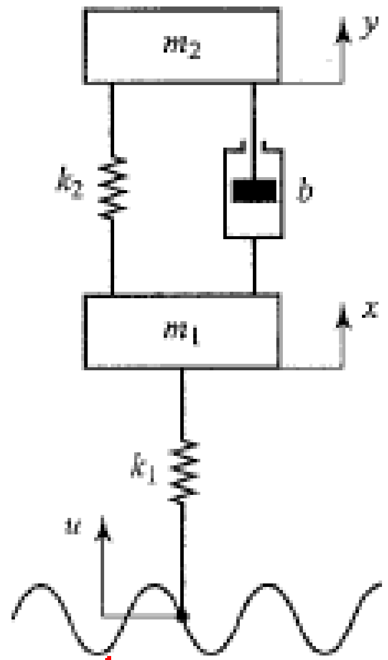
$$\frac{Y_1(s)}{F(s)} = \frac{1}{(M_1 s^2 + k_{12} + bs + k_1) - \frac{k_{12}^2}{M_2 s^2 + k_{12}}}$$

$\Rightarrow$ Order of the system = 4

= no. of storage elements (k_1, M_1, k_{12}, M_2)

Transfer Functions Examples – H.W

- Cause – u and Effect – displacement y , Find $\frac{Y(s)}{U(s)}$



$$m_2\ddot{y} + b(\dot{y}-\dot{x}) + k_2(y-x) = 0$$

$$m_1\ddot{x} + k_1(x-u) + k_2(x-y) + b(\dot{x}-\dot{y}) = 0$$

Apply LT with zero i c

$$(m_2s^2 + bs + k_2)Y(s) - (bs + k_2)X(s) = 0$$

$$(m_1s^2 + k_1 + k_2 + bs)X(s) - k_1U(s) - (k_2 + bs)Y(s) = 0$$

$$\begin{pmatrix} -(bs+k_2) & m_2s^2+bs+k_2 \\ m_1s^2+k_1+k_2+bs & -(bs+k_2) \end{pmatrix} \begin{pmatrix} X(s) \\ Y(s) \end{pmatrix} = \begin{pmatrix} 0 \\ +k_1 \end{pmatrix} U(s)$$

apply cramer's rule

$$Y(s) = \frac{\begin{vmatrix} -(bs+k_2) & 0 \\ (\;) & k_1 \end{vmatrix}}{\begin{vmatrix} & \\ & \end{vmatrix}} =$$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions Examples

- Cause – Force F and Effect – displacement x_2 $\quad \dfrac{X_2(s)}{F(s)}$

$$f(t) = M_1\frac{d^2x_1}{dt^2} + K_1x_1 + K_2(x_1 - x_2) + f_{v_3}\frac{d}{dt}(x_1 - x_2) + f_{v_1}\frac{dx_1}{dt}$$

$$0 = M_2\frac{d^2x_2}{dt^2} + K_3x_2 + K_2(x_2 - x_1) + f_{v_3}\frac{d}{dt}(x_2 - x_1) + f_{v_2}\frac{dx_2}{dt}$$

Apply LT with zero i.c

$$\left[M_1s^2(f_{v_1} + f_{v_3})s + (K_1 + K_2)\right]X_1(s) - (f_{v_3}s + K_2)X_2(s) = F(s)$$

$$-(f_{v_3}s + K_2)X_1(s) + \left[M_2s^2 + (f_{v_2} + f_{v_3})s + (K_2 + K_3)\right]X_2(s) = 0$$

$$\Delta = \begin{vmatrix} [M_1s^2 + (f_{v_1} + f_{v_3})s + (K_1 + K_2)] & -(f_{v_3}s + K_2) \\ -(f_{v_3}s + K_2) & [M_2s^2 + (f_{v_2} + f_{v_3})s + (K_2 + K_3)] \end{vmatrix}$$

$$X_2(s) = \frac{\begin{vmatrix} & F(s) \\ & 0 \end{vmatrix}}{\Delta}$$

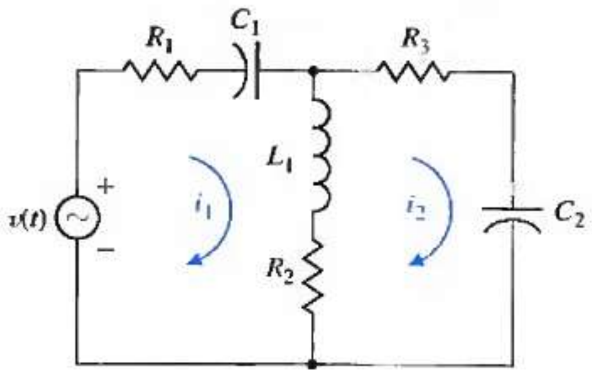
From this, the transfer function, $X_2(s)/F(s)$, is

$$\frac{X_2(s)}{F(s)} = G(s) = \frac{(f_{v_3}s + K_2)}{\Delta}$$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS
Transfer Functions Examples

- Cause – ~~e_1~~ $v(t)$ and Effect – ~~e_0~~ $i_2(t)$

find $\frac{I_2(s)}{V(s)}$



$$R_1 i_1 + \frac{1}{C_1}\int_{-\infty}^{t} i_1(\tau)\,d\tau + L_1 \frac{d}{dt}(i_1 - i_2) + R_2(i_1 - i_2) = v(t)$$

$$R_3 i_2 + \frac{1}{C_2}\int_{-\infty}^{t} i_2(\tau)\,d\tau + R_2(i_2 - i_1) + L_1 \frac{d}{dt}(i_2 - i_1) = 0$$

LT

$$\left(R_1 + \frac{1}{C_1 s} + L_1 s + R_2\right) I_1(s) - (L_1 s + R_2)\, I_2(s) = V(s)$$

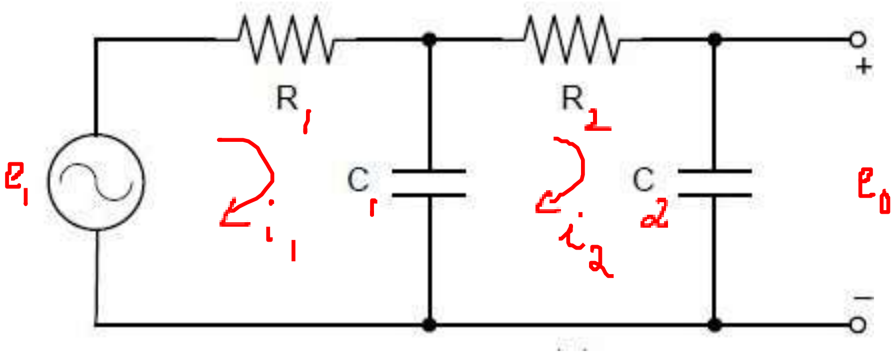
$$-(R_2 + L_1 s)\, I_1(s) + \left(R_3 + \frac{1}{C_2 s} + R_2 + L_1 s\right) I_2(s) = 0$$

$$I_2(s) = \frac{\begin{vmatrix} & V(s) \\ & 0 \end{vmatrix}}{\Delta}$$

$$\Delta = \begin{vmatrix} & \\ & \end{vmatrix}$$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS
Transfer Functions Examples

- Cause – e_1 and Effect – e_0 , Find $\frac{E_0(s)}{E_1(s)}$



$$\frac{1}{C_1}\int (i_1 - i_2)\,dt + R_1 i_1 = e_i$$

$$\frac{1}{C_1}\int (i_2 - i_1)\,dt + R_2 i_2 + \frac{1}{C_2}\int i_2\,dt = 0$$

$$\frac{1}{C_2}\int i_2\,dt = e_o$$

$$\frac{1}{C_1 s}[I_1(s) - I_2(s)] + R_1 I_1(s) = E_i(s)$$

$$\frac{1}{C_1 s}[I_2(s) - I_1(s)] + R_2 I_2(s) + \frac{1}{C_2 s} I_2(s) = 0$$

$$\frac{1}{C_2 s} I_2(s) = E_o(s)$$

$$\frac{E_o(s)}{E_i(s)} = \frac{1}{(R_1C_1s + 1)(R_2C_2s + 1) + R_1C_2s}$$

$$= \frac{1}{R_1C_1R_2C_2s^2 + (R_1C_1 + R_2C_2 + R_1C_2)s + 1}$$

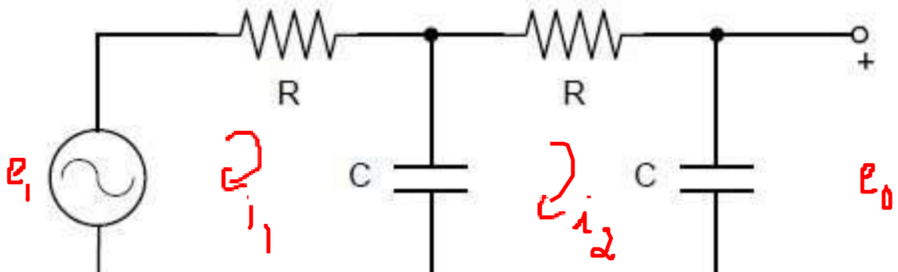
MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions Examples

- Cause – e_1 and Effect – e_0

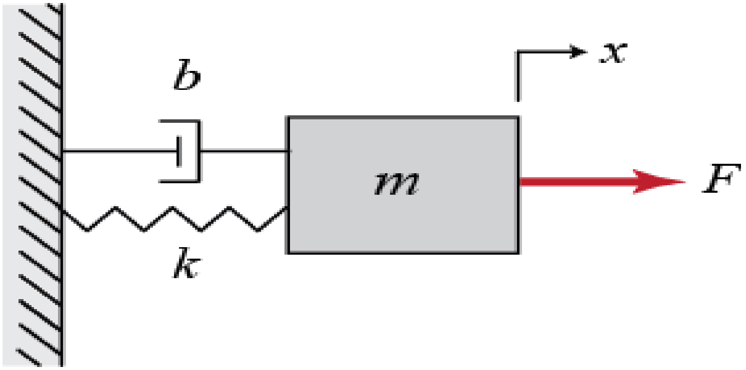
$$e_1 = R i_1 + \frac{1}{C}\int_{-\infty}^{t}(i_1 - i_2)\,d\tau$$

$$0 = R i_2 + \underbrace{\frac{1}{C}\int_{-\infty}^{t} i_2(\tau)\,d\tau}_{e_0} + \frac{1}{C}\int_{-\infty}^{t}(i_2 - i_1)\,d\tau$$

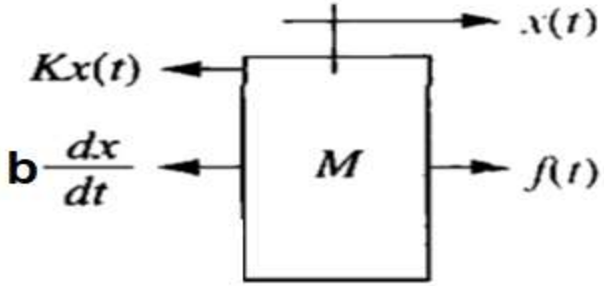


$$e_0(s) = \frac{I_2(s)}{Cs}$$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS
Transfer Functions Examples



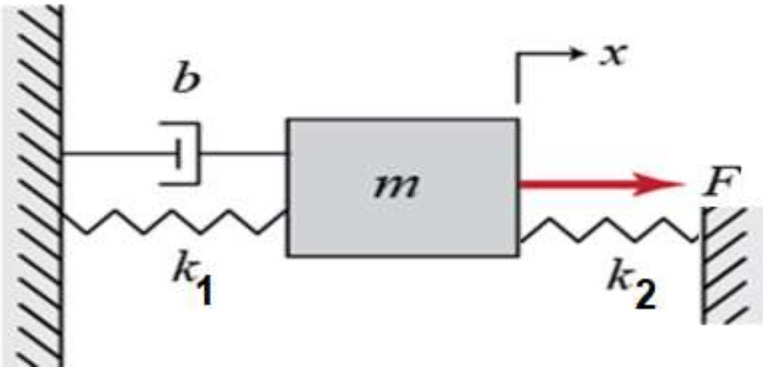
Free Body Diagram



$$f(t) - b\,\frac{dx(t)}{dt} - K\,x(t) = M\,\frac{d^2x(t)}{dt^2}$$

F(s) – bsX(s) - KX(s) = Ms^2X(s)

T(s) = $\frac{X(s)}{F(s)} = \frac{1}{Ms^2+bs+K}$



MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Analogous System Examples

F-V

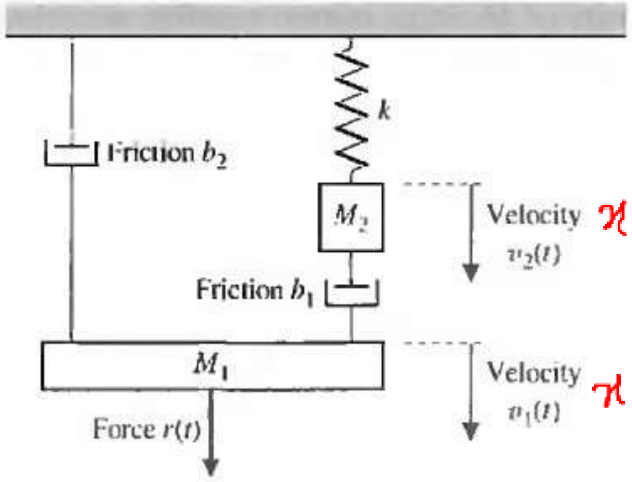
$M = L$, $b = R$, $k = \frac{1}{C}$

F-I, $M = C$, $b = \frac{1}{R}$, $k = \frac{1}{L}$

$$r(t) = M_1 \ddot{x}_1 + b_2 \frac{dx_1}{dt} + b_1 \frac{d(x_1 - x_2)}{dt}$$

$$= M_1 \frac{dv_1}{dt} + b_2 v_1 + b_1 (v_1 - v_2)$$

$$0 = M_2 \ddot{x}_2 + b_1(\dot{x}_2 - \dot{x}_1) + k x_2$$

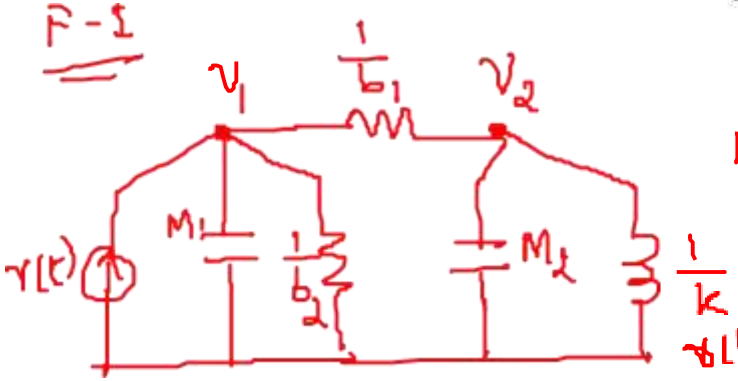


$x_2 \Rightarrow i_2 \quad V_2$

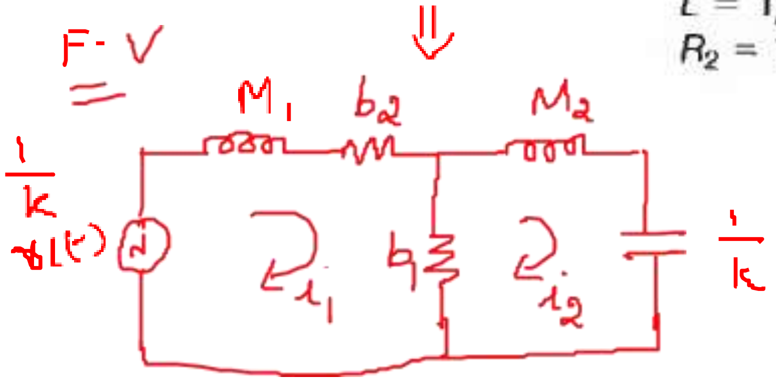
$x_1 \Rightarrow i_1 \quad v_1$

$C_1 = M_1$, $C_2 = M_2$,
$L = 1/k$, $R_1 = 1/b_1$,
$R_2 = 1/b_2$.

F-I



F-V



MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Analogous System Examples

$$b_1(\dot{x}_i - \dot{x}_o) + k_1(x_i - x_o) = b_2(\dot{x}_o - \dot{y})$$

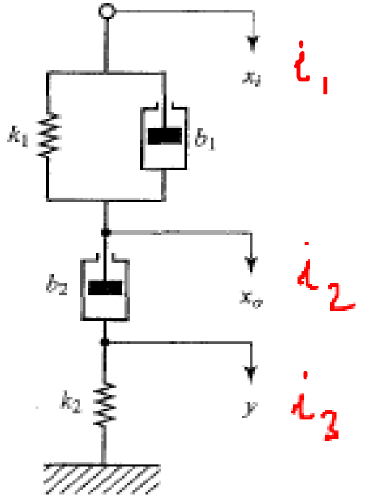
$$b_2(\dot{x}_o - \dot{y}) = k_2 y$$

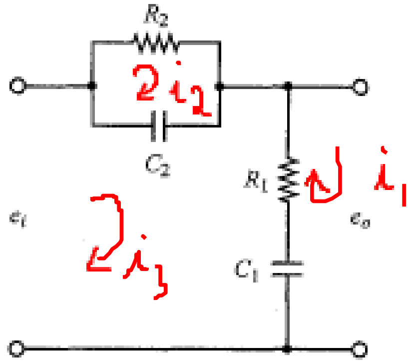
$$b_1[sX_i(s) - sX_o(s)] + k_1[X_i(s) - X_o(s)] = \ldots$$

$$b_1[sX_i(s) - sX_o(s)] + k_1[X_i(s) - X_o(s)] = b_2 sX_o(s) - b_2 s\frac{b_2 sX_o(s)}{b_2 s + k_2}$$

$$(b_1 s + k_1)X_i(s) = \left(b_1 s + k_1 + b_2 s - b_2 s\frac{b_2 s}{b_2 s + k_2}\right)X_o(s)$$

$$\frac{X_o(s)}{X_i(s)} = \frac{\left(\frac{b_1}{k_1}s + 1\right)\left(\frac{b_2}{k_2}s + 1\right)}{\left(\frac{b_1}{k_1}s + 1\right)\left(\frac{b_2}{k_2}s + 1\right) + \frac{b_2}{k_1}s}$$

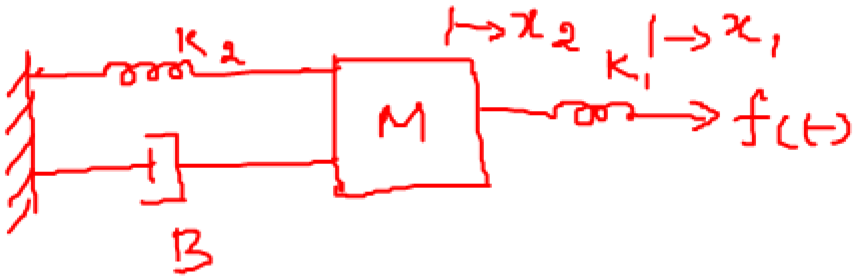




MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Analogous System Examples

1)

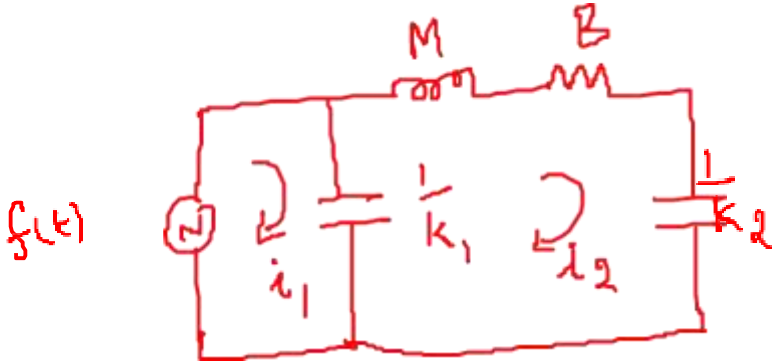


$$0 = M\frac{d^2x_2}{dt^2} + K_2x_2 + B\frac{dx_2}{dt} + \underbrace{K_1(x_2 - x_1)}$$

$$f(t) = K_1(x_1 - x_2)$$

f - V

$M = L$, $B = R$, $K = 1/C$

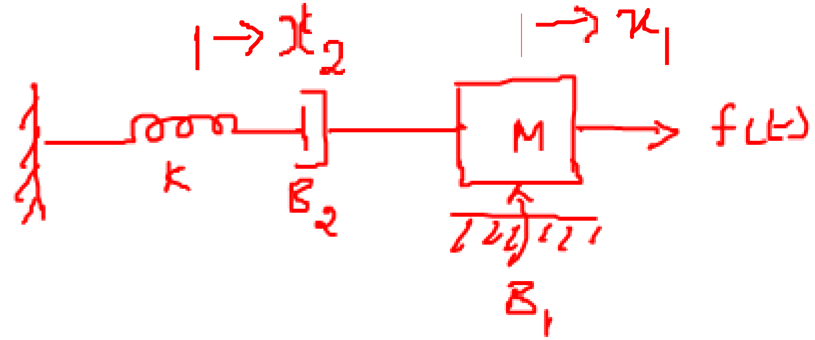


f - I ⇒ $M = C$, $B = \frac{1}{R}$, $K = \frac{1}{L}$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions Examples

2)



$$f(t) = M\frac{d^2x_1}{dt^2} + B_2\frac{d(x_1-x_2)}{dt} + B_1\frac{dx_1}{dt}$$

$$0 = Kx_2 + B_2\frac{d(x_2-x_1)}{dt}$$

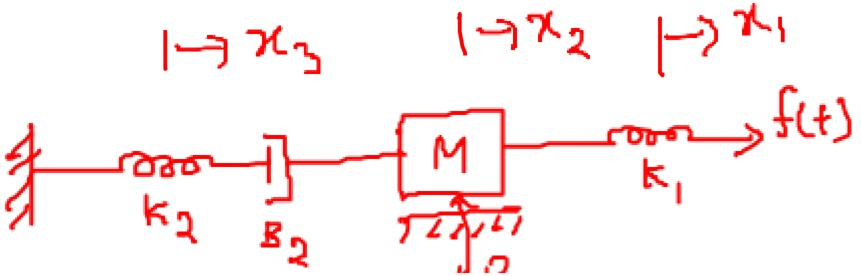
f-$V \Rightarrow M = L,\ B = R,\ K = \frac{1}{C}$

f-$I \Rightarrow M = C,\ B = \frac{1}{R},\ K = \frac{1}{L}$

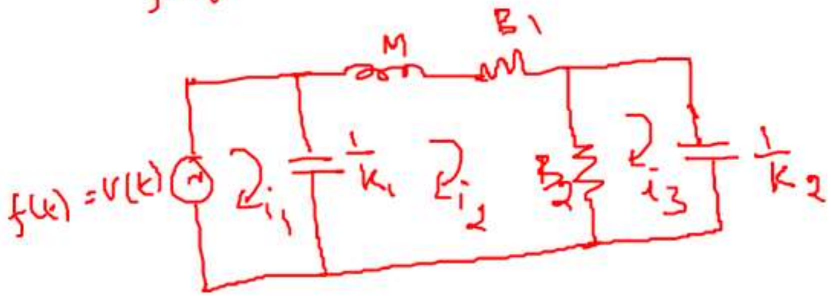
MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Analogous System Examples

3)



f - V



$$f(t) = K_1(x_1 - x_2)$$

$$0 = M\frac{d^2x_2}{dt^2} + B_1\frac{dx_2}{dt} + B_2\frac{d(x_2 - x_3)}{dt} + K_1(x_2 - x_1)$$

$$0 = K_2x_3 + B_2\frac{d(x_3 - x_2)}{dt}$$

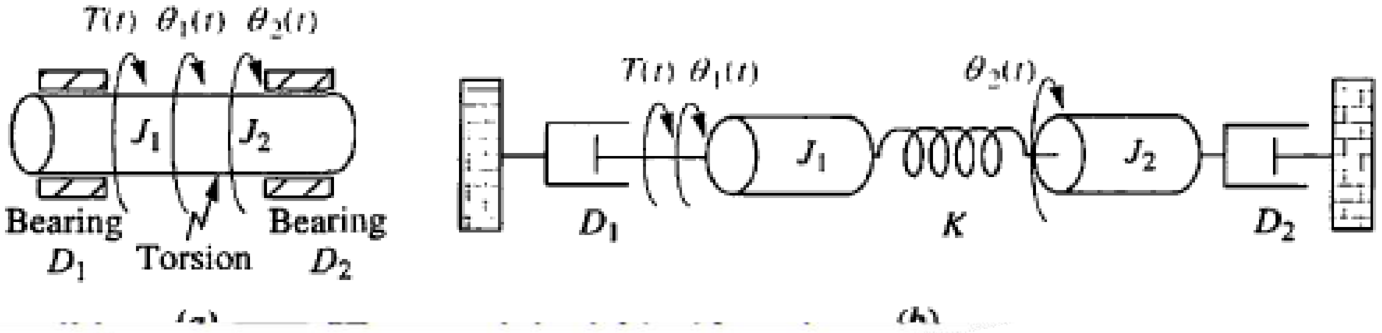
f - I ~ H.W

$$\begin{bmatrix} J_1s^2+D_1s+K & -K \\ -K & J_2s^2+D_2s+K \end{bmatrix}\begin{pmatrix}\theta_1 \\ \theta_2\end{pmatrix} =$$

- Cause – T and Effect – θ_2

$$T(t) = J_1\frac{d^2\theta_1}{dt^2} + D_1\frac{d\theta_1}{dt} + K(\theta_1-\theta_2)$$

$$0 = J_2\frac{d^2\theta_2}{dt^2} + D_2\frac{d\theta_2}{dt} + K(\theta_2-\theta_1)$$



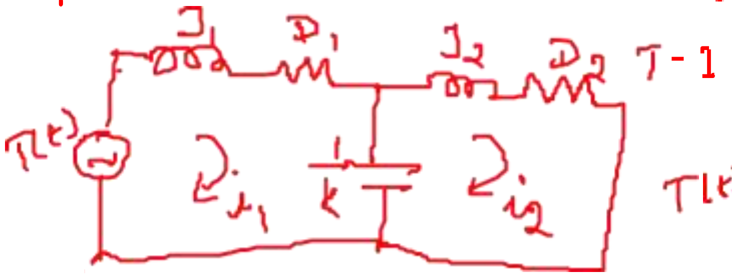
$$(J_1s^2+D_1s+K)\theta_1(s) \qquad -K\theta_2(s)$$

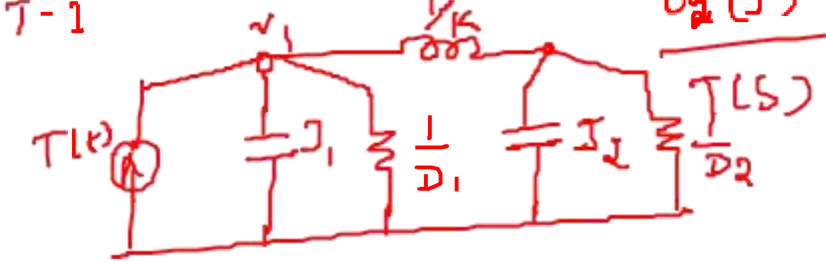
$$\theta_2(s) = \frac{\begin{vmatrix} J_1s^2+D_1s+K & T(s) \\ -K & 0 \end{vmatrix}}{\Delta}$$

$$\frac{\theta_2(s)}{T(s)} \quad K$$

T- V

T - I





$$\frac{\theta_2(s)}{T(s)} = \frac{K}{(J_1s^2+D_1s+K)(J_2s^2+D_2s+K) - K^2}$$

$$\Delta = \begin{vmatrix} (J_1s^2+D_1s+K) & -K \\ -K & (J_2s^2+D_2s+K) \end{vmatrix}$$

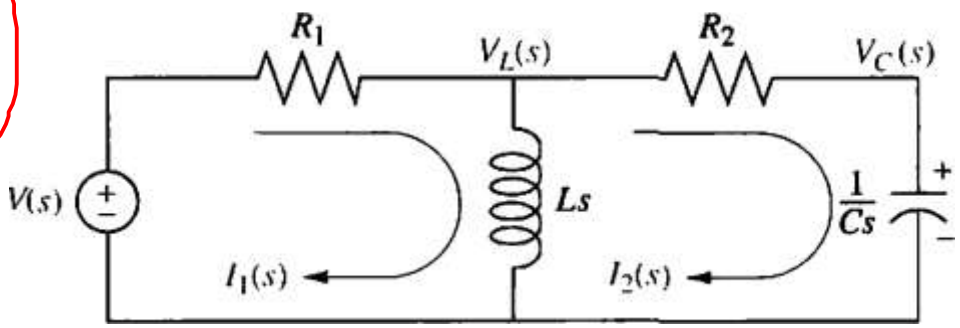
Transfer Functions Examples

- Cause –v(t) and Effect – $i_2(t)$

$$R_1 I_1(s) + LsI_1(s) - LsI_2(s) = V(s)$$

$$LsI_2(s) + R_2 I_2(s) + \frac{1}{Cs} I_2(s) - LsI_1(s) = 0$$

$$\begin{bmatrix} R_1 + Ls & -Ls \\ -Ls & Ls + R_2 + \frac{1}{Cs} \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} V(s) \\ 0 \end{bmatrix}$$



$$V(s) \longrightarrow \boxed{\frac{LCs^2}{(R_1 + R_2)LCs^2 + (R_1 R_2 C + L)s + R_1}} \longrightarrow I_2(s)$$

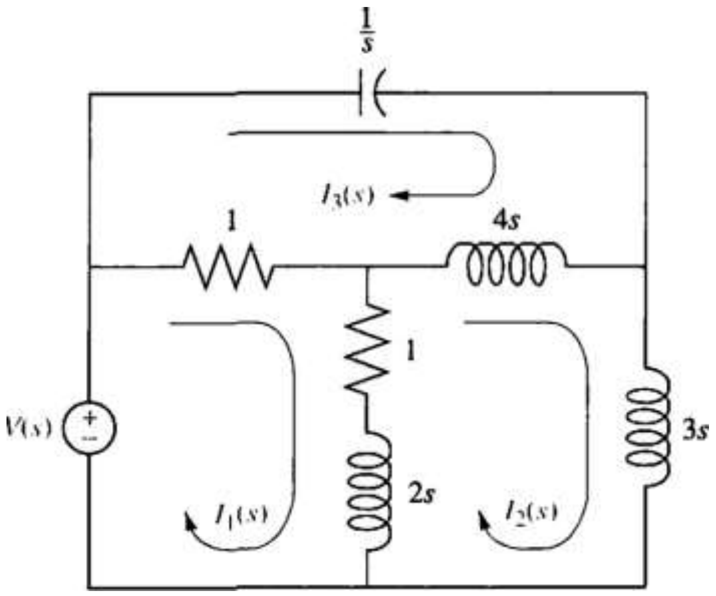
MATHEMATICAL MODELS OF PHYSICAL SYSTEMS
Transfer Functions Examples – H.W

- Cause –v(t) and Effect – $i_2(t)$

$$+(2s+2)I_1(s) - (2s+1)I_2(s) \qquad - I_3(s) = V(s)$$

$$-(2s+1)I_1(s) + (9s+1)I_2(s) \qquad - 4sI_3(s) = 0$$

$$-I_1(s) \qquad - 4sI_2(s) + \left(4s+1+\tfrac{1}{s}\right)I_3(s) = 0$$



THANK YOU

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